

Fix-the-Design Question Bank (Detailed Model Fixes)

Q1) FIX for “Elevator Panel Confusion” (Visibility + Mapping + Signifiers)

Broken design: Elevator panel has labels that look like buttons. Users press labels mistakenly (the classic bad elevator example).

Task: Fix the design using **Visibility + Mapping** (and mention **signifiers / affordances**).

Model fix (what you would write in the exam):

This design fails because the **visibility of controls** is weak: users cannot immediately tell which elements are pressable, so labels are mistaken for buttons (poor signifiers). Visibility means relevant actions must be obvious at the moment of need . It also has poor **mapping** because the relationship between label and button is not visually adjacent or clearly grouped; mapping should make control–effect relationships logical and natural .

Redesign changes:

1. **Separate labels from controls clearly:** Put labels **above** buttons, not beside them; labels in smaller, non-button styling (plain text).
2. **Make buttons look like buttons (strong signifiers):** Use raised/outlined circular buttons with tactile cues (bevel, shadow) so they **afford pressing** .
3. **Use grouping + proximity:** Group floors in a clean grid (e.g., 1–12), with consistent spacing and alignment.
4. **Add feedback for presses:** When pressed, button lights up + click sound (feedback is “system tells user what happened”) .
5. **Error prevention constraint:** Disable impossible options (e.g., floors not available) instead of showing them as normal. Constraints prevent wrong actions by limiting what can be done .

Before/after outcome: Users instantly recognize “these circles are buttons,” and because label–button are visually tied, errors drop and speed increases.

Q2) FIX for “Vending machine expects button press before money” (Mental Model + Consistency + Guidance)

Broken design: Vending machine requires “Select item” before inserting money; users try money first and think it’s broken.

Task: Fix using **Match to real world/mental model**, and add **visibility + feedback**.

Model fix:

The main issue is mismatch with the user’s **mental model** and real-world convention: users expect the common sequence “insert money → select item.” If you break a strong convention, you must redesign carefully with guidance (otherwise users assume the system failed). This is directly tied to “match between system and the real world” (Nielsen) and to **consistency** (users transfer habits from similar systems) .

Redesign option A (best): Align to convention

1. Default state: screen shows “**Insert cash/card to begin**” with visible money slot lighting (visibility).
2. When money inserted: provide immediate **feedback** (“Rs 100 credited”) and unlock item buttons.
3. After selection: show “Dispensing...” progress feedback and final closure.

Redesign option B (if they *must* keep select-first): Add strong guidance

1. Bold instruction at top: “Step 1: Select item (buttons will light up)”
2. Make buttons flash subtly to suggest required action (visibility + signifier)
3. After selection, money slot lights up + screen says “Step 2: Insert money now” (feedback + guidance)

Outcome: The system becomes learnable because the flow matches expectations or teaches the non-standard flow explicitly.

Q3) FIX for “Remote control triangles: Volume vs Channel” (Distinguish controls + Prevent slips)

Broken design: Volume and channel are both identical up/down triangles; only tiny labels differentiate.

Task: Fix using **visibility**, **mapping**, and “prevent errors/slips.”

Model fix:

This UI causes **slips** because the two controls are perceptually too similar. Visibility requires that important controls be obvious and easily distinguishable . Mapping should also clearly show which arrows correspond to volume vs channel .

Redesign changes:

1. **Distinct shapes/size:** Make Volume rocker long horizontal with “+ / –” and speaker icon; Channel rocker vertical with “▲/▼” and “CH”.
2. **Spatial mapping:** Place volume near audio-related controls; place channel near numeric keypad.
3. **Tactile differentiation:** Different texture or ridge so users can feel it without looking (supports eyes-free use).
4. **Feedback:** On-screen overlay “Volume: 12” vs “Channel: 45” confirms which action occurred (feedback) .

Outcome: Faster operation + fewer accidental channel changes.

Q4) FIX for “Calendar where decoration hides dates” (SNR + Minimalism + Task priority)

Broken design: Calendar month view has huge decorative graphics that cover date numbers.

Task: Redesign using “function first,” **signal-to-noise** / minimalism, and usability.

Model fix:

The primary task of a calendar is “see dates quickly.” Decoration that blocks dates reduces **signal-to-noise ratio** and violates minimalist design principles; the interface becomes visually noisy and harms efficiency. Minimal design is aligned with “aesthetic and minimalist design” (Nielsen) .

Redesign changes:

1. **Clear grid first:** High-contrast date numbers, consistent alignment, enough whitespace.
2. **Decoration as background only:** Put imagery in a low-opacity background that never intersects numbers.
3. **Hierarchy:** Weekdays header, date cells, then events; decoration is last priority.
4. **Accessibility:** Ensure contrast works, not color-only meaning.
5. **Optional theme toggle:** Let users enable/disable decorative mode.

Outcome: Usability restored while still allowing aesthetics.

Q5) FIX for “Card reader slot not obvious” (Visibility + Signifier + Feedback)

Broken design: Hotel elevator requires room card, but card slot is not visible/obvious.

Task: Fix using **visibility** and **feedback**.

Model fix:

This fails **visibility** because the user cannot see what action is required or where to do it. Visibility means the system should make relevant parts/actions obvious . It also lacks feedback: when the user tries, there's no confirmation. Feedback is "system sends info back after action" .

Redesign changes:

1. **Make slot visually prominent:** Place at eye-level with strong border, icon, label "Tap/Insert card here."
2. **Use lighting as signifier:** LED ring around slot that pulses gently when card required.
3. **Audio/haptic feedback:** Beep + green light + message "Access granted."
4. **Constraint:** Disable floor buttons until card recognized (constraint prevents wrong action) .
5. **Error message clarity:** If rejected, show "Invalid card—try again or contact reception" (recoverable).

Outcome: Users understand the required step instantly, and confirmation reduces anxiety.

Q6) FIX for "Greyed-out Submit with no reason" (Constraints + Feedback + Error recovery)

Broken design: Submit button is disabled but user doesn't know why.

Task: Redesign using **constraints** + **feedback** + "help users recover."

Model fix:

Constraints are valid (disabling prevents invalid submission) but **constraints must be explainable**. Constraints restrict actions to prevent errors , while feedback must communicate what the system needs from the user . Without explanation, the user is stuck.

Redesign changes:

1. Keep submit disabled (good constraint) but show **inline checklist**:
 - "Upload CNIC front (missing)"
 - "Enter phone number (missing)"
2. Highlight missing fields with text + icon (not only color).
3. When user taps disabled submit, show tooltip: "Complete required fields to continue."
4. Provide validation in real-time (as user types).
5. Place error messages near the field, in plain language (support recovery).

Outcome: Button state becomes informative, not confusing.

Q7) FIX for “Ambiguous road sign: do not enter + right arrow” (Clarity + Unambiguous communication)

Broken design: Road sign shows “do not enter” and an arrow together (conflicting message).

Task: Fix the sign using clarity and error prevention.

Model fix:

This is a classic **ambiguity** problem: the sign communicates two conflicting actions. In interaction design, communication must be **clear and unambiguous**, especially in high-risk contexts (error consequences are severe). This is also error prevention: preventing wrong actions through clear constraints/messages .

Redesign changes:

1. Split into two signs:
 - “No entry” placed directly at blocked lane
 - “Detour →” placed before decision point
2. Use consistent standard symbols (external consistency).
3. Put arrow sign earlier so driver has time to react.
4. Increase legibility: large arrow, high contrast, minimal text.

Outcome: Removes confusion, reduces accidents.

Q8) FIX for “Phone keypad vs calculator layout confusion” (External Consistency + Standard)

Broken design: Calculator app uses phone keypad layout (1–2–3 top), confusing users.

Task: Redesign using **external consistency**.

Model fix:

External consistency means similar systems follow the same conventions so skills transfer. Your notes explicitly mention the phone vs calculator keypad mismatch as a real example of external inconsistency . For a calculator, the dominant convention is 7–8–9 on top.

Redesign changes:

1. Use standard calculator layout: 7–8–9 top row.
2. Keep operations (+ – × ÷) in consistent right column.
3. Separate “C/AC” and “=” with visual emphasis to prevent destructive slips.
4. If the app must support both, provide a settings toggle “Phone layout / Calculator layout,” but default to the standard.

Outcome: Faster entry, fewer errors.

Q9) FIX for “Touch target too small on mobile” (Fitts’ Law + Mobile constraints)

Broken design: On mobile checkout, the “Pay” button is a tiny text link in the corner.

Task: Fix using mobile UX constraints + easy touch.

Model fix:

On mobile, users are handheld and often one-handed; touch targets must be reachable and large enough. (Mobile users are constrained physically; interaction must support thumb use.) Small targets increase errors and time.

Redesign changes:

1. Make “Pay” a primary bottom CTA (full-width, large height).
2. Place it in thumb-friendly zone (lower half).
3. Use clear label “Pay Rs ____” for visibility of outcome.
4. Provide feedback state: pressed/processing/success.
5. Ensure disabled states show reason (combine with constraints).

Outcome: Higher success rate, fewer mis-taps, better efficiency.

Q10) FIX for “Too many options on home screen” (Hick’s Law + Grouping + Progressive disclosure)

Broken design: Home page shows 14 equal buttons (Ride, Food, Courier, Grocery, etc.).

Task: Redesign using decision simplification.

Model fix:

Too many equal choices increases decision time (classic choice overload). Aesthetic/minimal design expects only the most relevant info visible, and reduce memory/attention demands.

Redesign changes:

1. Identify top 3–4 frequent actions (80/20 principle) and show them as primary cards.
2. Group remaining into “More services” with categories.
3. Add search bar (“What do you want to do?”) to support expert users.
4. Use whitespace and alignment for clean scanning.

5. Maintain consistency of icons and labels across app.

Outcome: Faster decisions, cleaner UX.

Q11) FIX for “No closure after payment” (Dialog closure + Feedback)

Broken design: Payment completes and app jumps to home with no receipt/success screen.

Task: Fix using **closure** and feedback.

Model fix:

Shneiderman explicitly says “**Design dialogs to yield closure**” and “Offer informative feedback.” Users need a completion state that confirms success, summarizes what happened, and provides next steps.

Redesign changes:

1. Add success confirmation screen: “Payment successful.”
2. Provide summary: amount, merchant, date/time, reference ID.
3. Provide next actions: “Download receipt,” “View order,” “Return home.”
4. Send receipt via email/SMS as redundancy.
5. Provide error alternative flow if failed: show reason + recovery actions.

Outcome: User trust increases, anxiety decreases, fewer support complaints.

Q12) FIX for “Color-only status indicator (green/red)” (Accessibility + Redundancy)

Broken design: Dashboard uses only green vs red dots for status.

Task: Fix using accessibility and inclusive design.

Model fix:

Using color alone violates accessibility because color-blind users may not distinguish red/green. Your notes emphasize not assuming everyone sees color the same way . Inclusive design requires redundant cues.

Redesign changes:

1. Add text labels (“Active / Inactive”).

2. Add icons (✓ / ! / ×).
3. Use patterns or shapes (filled circle vs triangle).
4. Ensure contrast ratio is strong; don't rely on subtle shades.
5. Provide tooltips for clarity.

Outcome: Everyone can interpret status correctly.

How to Answer Fix-the-Design in Your Sir's Style (1 template you can reuse)

When you write, use this order every time:

1. **Name the principle** (visibility/feedback/constraints/mapping/consistency/affordances)
2. **Explain what is broken in this design** (1–2 lines)
3. **Redesign** (3–6 concrete UI changes)
4. **Why it works** (ties back to principle + user errors avoided + speed improved)